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Q1. (Code at the end of the document, Pg 7 onwards)

Using the same fuzzy rule base for all three systems (low $\rightarrow$ small tip, average $\rightarrow$ medium tip, high $\rightarrow$ large tip), the difference in outcomes comes purely from the shape of the membership functions: triangular, trapezoidal, and Gaussian.

With triangular membership functions, the transitions between linguistic terms are linear and symmetric. This produces smooth, balanced rule activations. As seen in Test Case 1 and 3, triangular functions allow partial activation of neighboring rules (e.g., $R1 = 0.800$ and $R2 = 0.200$ for poor inputs), leading to moderate tip outputs such as 7.73% and 20.10%. The results are generally stable and intuitive, sitting between the more extreme trapezoidal and smoother Gaussian outputs.

Trapezoidal membership functions create flat “plateaus” where membership is fully 1 over a range of inputs. This makes the system more decisive when inputs clearly fall into a category. For example, in Test Case 1 (very poor inputs), $R1 = 1.000$ and the tip drops sharply to 3.82%, the lowest among all methods. Similarly, in Test Case 3 (excellent inputs), $R3 = 1.000$ and the tip rises to 22.35%, the highest result. Thus, trapezoidal functions tend to produce more extreme and crisp decisions.

Gaussian membership functions provide smooth, bell-shaped curves with gradual transitions. They allow significant overlap between categories, so multiple rules are often activated simultaneously (e.g., in Test Case 6, $R2 = 0.607$ and $R3 = 0.801$). This produces softer, more blended decisions such as 17.91%. Gaussian outputs are typically less abrupt than trapezoidal and slightly smoother than triangular, often landing between the two or gently shifting toward the dominant category.

Overall, all three methods follow the same logical structure and produce similar trends: low inputs yield low tips, average inputs yield mid-range tips (~14 to 15%), and high inputs yield high tips (~20 to 22%). The main difference lies in sensitivity and decisiveness. Trapezoidal functions are the most decisive and produce stronger extremes; triangular functions provide balanced linear transitions, and Gaussian functions create the smoothest and most gradual blending between rules.

Q2. A)

I). For $\gamma = 1$ there is no penalty for taking longer paths, so the agent only cares about the final reward magnitude, not how far away it is. Since exiting at A gives reward 10 and exiting at E gives reward 1, the optimal strategy everywhere is to head toward A and collect the larger reward.

Thus, the optimal policy is at A $\rightarrow$ Exit; at B $\rightarrow$ West; at C $\rightarrow$ West; at D $\rightarrow$ West; at E $\rightarrow$ West.

Even in state E it is better to move West and travel to A for reward 10 than to exit immediately for reward 1, because with $\gamma = 1$ future rewards are not discounted.

II). For $\gamma = 0.1$ the agent strongly prefers immediate rewards, so distance matters much more than reward size. We compare discounted returns to each exit.

From B:

to A $\rightarrow 110 \cdot 0.1 = 11$, to E $\rightarrow 1 \cdot 0.1 = 0.1 \Rightarrow$ go West.

From C:

to A $\rightarrow 110 \cdot 0.12 = 0.1$, to E $\rightarrow 11 \cdot 0.12 = 0.01 \Rightarrow$ go West.

From D:

to A $\rightarrow 110 \cdot 0.13 = 0.01$, to E $\rightarrow 1 \cdot 0.1 = 0.1 \Rightarrow$ go East.

At A and E you exit immediately.

The optimal policy for $\gamma = 0.1$ is:

A $\rightarrow$ Exit, B $\rightarrow$ West, C $\rightarrow$ West, D $\rightarrow$ East, E $\rightarrow$ Exit.

III). In state D we make West and East equally good.

Going West from D reaches A in 3 steps: value = $10\gamma^3$.

Going East from D reaches E in 1 step: value = γ .

Set them equal:

$$10\gamma^3 = \gamma$$

$$10\gamma^2 = 1$$

$$\gamma^2 = 0.1$$

$$\gamma = \sqrt{0.1}$$

$$\gamma \approx 0.316$$

So, West and East are equally good in state D when $\gamma \approx 0.316$.

B)

We interpret the diagram as an MDP with 3 states:

Cool (C), Warm (W), Overheated (O, terminal)

Actions: Slow, Fast

Discount: not shown $\rightarrow$ assume $\gamma = 1$

Terminal state: $V(O) = 0$

Transitions and rewards from the diagram:

From Cool

Slow $\rightarrow$ C with prob 1.0, reward +1

Fast $\rightarrow$ C with prob 0.5, reward +2

Fast $\rightarrow$ W with prob 0.5, reward +2

From Warm

Slow $\rightarrow$ C with prob 0.5, reward +1

Slow $\rightarrow$ W with prob 0.5, reward +1

Fast $\rightarrow$ O with prob 1.0, reward -10

Initialize $V_0(C) = V_0(W) = V_0(O) = 0$

$k = 1$

$$Q_1(C, \text{Slow}) = 1$$

$$Q_1(C, \text{Fast}) = 2$$

$$V_1(C) = 2$$

$$Q_1(W, \text{Slow}) = 1$$

$$Q_1(W, \text{Fast}) = -10$$

$$V_1(W) = 1$$

$$\mathbf{k = 2}$$

$$Q_2(C, \text{Slow}) = 1 + V_1(C) = 3$$

$$Q_2(C, \text{Fast}) = 2 + 0.5 \cdot V_1(C) + 0.5 \cdot V_1(W)$$

$$= 2 + 1 + 0.5 = 3.5$$

$$V_2(C) = 3.5$$

$$Q_2(W, \text{Slow}) = 1 + 0.5 \cdot V_1(C) + 0.5 \cdot V_1(W)$$

$$= 1 + 1 + 0.5 = 2.5$$

$$Q_2(W, \text{Fast}) = -10$$

$$V_2(W) = 2.5$$

$$\mathbf{k = 3}$$

$$Q_3(C, \text{Slow}) = 1 + 3.5 = 4.5$$

$$Q_3(C, \text{Fast}) = 2 + 0.5 \cdot 3.5 + 0.5 \cdot 2.5$$

$$= 2 + 1.75 + 1.25 = 5$$

$$V_3(C) = 5$$

$$Q_3(W, \text{Slow}) = 1 + 0.5 \cdot 3.5 + 0.5 \cdot 2.5$$

$$= 1 + 1.75 + 1.25 = 4$$

$$Q_3(W, \text{Fast}) = -10$$

$$V_3(W) = 4$$

$$\mathbf{k = 4}$$

$$Q_4(C, \text{Slow}) = 1 + 5 = 6$$

$$Q_4(C, \text{Fast}) = 2 + 0.5 \cdot 5 + 0.5 \cdot 4$$

$$= 2 + 2.5 + 2 = 6.5$$

$$V_4(C) = 6.5$$

$$Q_4(W, \text{Slow}) = 1 + 0.5 \cdot 5 + 0.5 \cdot 4$$

$$= 1 + 2.5 + 2 = 5.5$$

$$Q_4(W, \text{Fast}) = -10$$

$$V_4(W) = 5.5$$

$$V_4(O) = 0$$

Final values (k=4)

$$V_4(C)=6.5$$

$$V_4(W)=5.5$$

$$V_4(O)=0$$

$$Q_4(C,Slow)=6$$

$$Q_4(C,Fast)=6.5$$

$$Q_4(W,Slow)=5.5$$

$$Q_4(W,Fast)=-10$$

Suggested policy

In Cool: choose Fast

In Warm: choose Slow

Overheated: terminal

The optimal policy is: drive Fast when cool, switch to Slow when warm to avoid overheating.

Q5.

Using the Bayesian network factorization:

$$P(b,e,a,j,m)=P(b)P(e)P(a|b,e)P(j|a)P(m|a)$$

i) $P(+b, -e, +a, -j, +m)$

$$=0.01 \times 0.98 \times 0.95 \times 0.2 \times 0.7 = 0.0013034$$

- This represents: burglary occurs, no earthquake, alarm rings, John does not call, Mary calls.

ii) $P(+b, +e, +a, -j, +m)$

$$=0.01 \times 0.02 \times 0.9 \times 0.2 \times 0.7 = 0.0000252$$

- This represents: burglary and earthquake occur, alarm rings, John does not call, Mary calls.

iii) $P(+b, +e, +a, +j, +m)$

$$=0.01 \times 0.02 \times 0.9 \times 0.8 \times 0.7 = 0.0001008$$

- This represents: burglary and earthquake occur, alarm rings, both John and Mary call.

iv) $P(-b, +e, +a, -j, +m)$

$$=0.99 \times 0.02 \times 0.3 \times 0.2 \times 0.7 = 0.0008316$$

- This represents: no burglary, earthquake occurs, alarm rings, John does not call, Mary calls.

v) $P(-b, -e, -a, -j, -m)$

$$=0.99 \times 0.98 \times 0.98 \times 0.9 \times 0.95 = 0.81293$$

- This represents: no burglary, no earthquake, no alarm, and neither John nor Mary calls.

The joint probability is computed by multiplying local conditional probabilities because the Bayesian network encodes conditional independence. Burglary and earthquake are independent causes, alarm depends only on them, and John/Mary depend only on the alarm. This structure allows the full joint distribution to be factorized into smaller components.

The Bayesian network simplifies joint probability computation by exploiting conditional independence between variables. Instead of using one large table, we multiply smaller conditional probabilities derived from the network structure. This makes reasoning about uncertain events efficient and mathematically consistent.